Requirement Analysis

1.Data flow diagram:

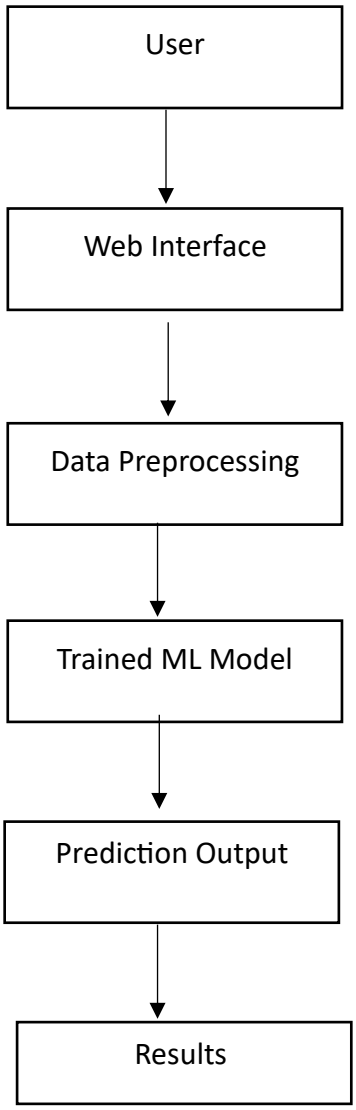
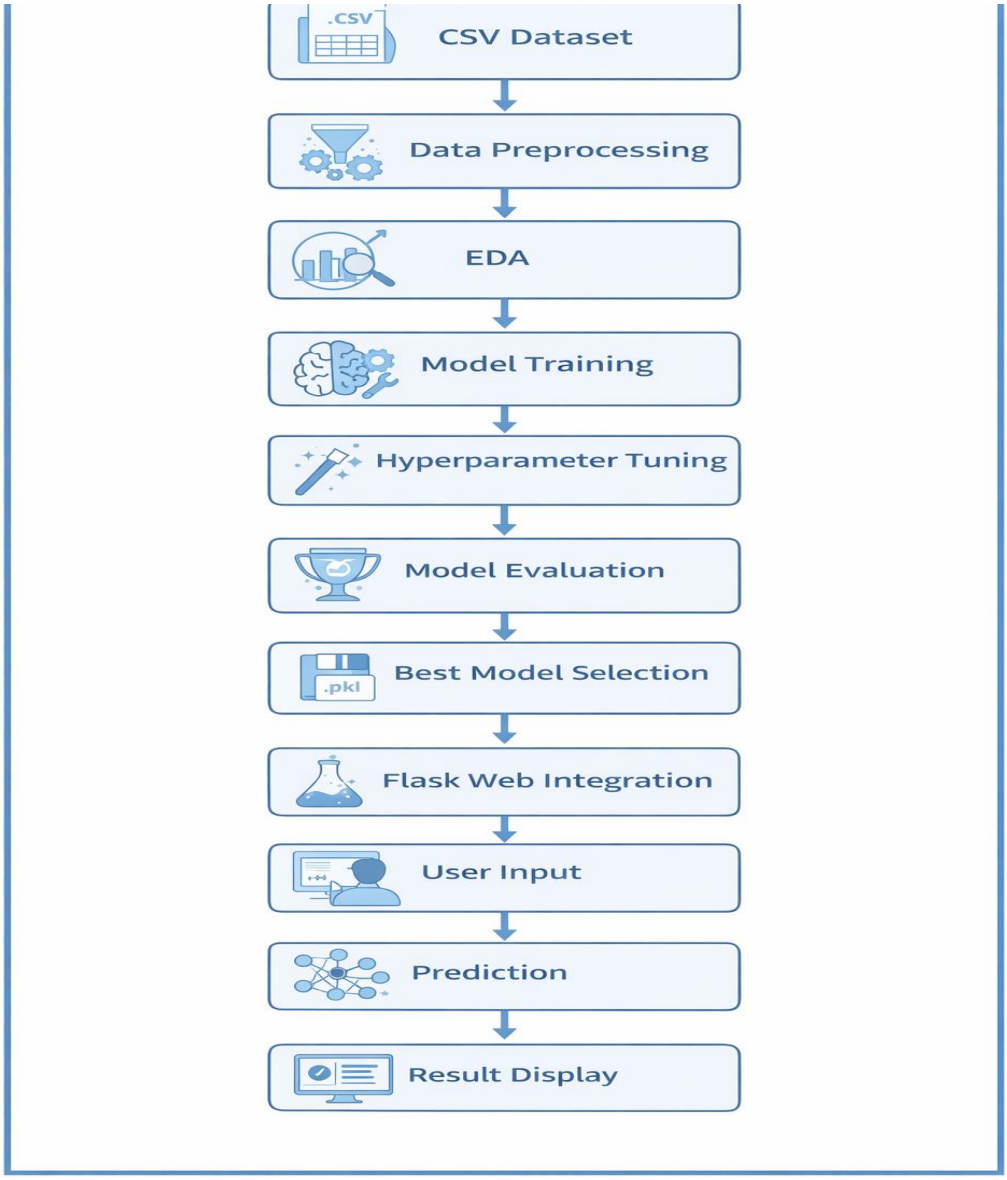


Fig-1: Project Architecture

2.solution requirements:

The system must accept the startup dataset in CSV format and perform proper data handling procedures such as managing missing values, encoding categorical variables, and scaling numerical features. The dataset must be divided into training and testing sets to ensure proper model validation. The system should train multiple machine learning algorithms including Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine. Each model must be evaluated using performance metrics such as Accuracy, Precision,

Recall, F1-Score, and Confusion Matrix. The system should compare the performance of all algorithms and apply hyperparameter tuning to improve model efficiency, followed by a comparison of results before and after tuning. Based on balanced evaluation metrics, the best-performing model must be selected and saved as a .pkl file for deployment. The web application should include a Home page, Prediction page, and Result page, allowing users to input startup details and receive a Success or Failure prediction within seconds. The system must validate user inputs, ensure security, maintain scalability for multiple users, and support deployment using the Flask framework either on localhost or cloud platforms.

3. Technology stocks: